CERTIFICATE OF EXCELLENCE

THIS CERTIFICATE IS AWARDED TO

SCALER Topics

Sahith M

In recognition of the completion of the tutorial: **DBMS Course - Master the Fundamentals and Advanced Concepts**

Following are the the learning items, which are covered in this tutorial

74 Video Tutorials 16 Modules 16 Challenges

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During my study of relational databases, I learned how data can be organized efficiently using the **Relational Database Model** and visualized with **Entity-Relationship (ER) diagrams**, which clarify the relationships between entities and attributes. I gained practical skills in **SQL**, including writing queries with `SELECT`, `JOIN`, `GROUP BY`, subqueries, and applying DML operations like `INSERT`, `UPDATE`, and `DELETE`. I also explored **indexing techniques**, such as B-trees and B+ trees, which improve data retrieval speed, and **normalization**, which eliminates redundancy and ensures data integrity by enforcing functional dependencies across different normal forms.

Two key takeaways are:

1. **Normalization is critical**—understanding 1NF, 2NF, 3NF, BCNF, and beyond helps maintain data consistency and reduces anomalies.
2. **Indexes dramatically improve performance**, especially for large datasets, by allowing fast lookup and range queries without scanning the entire table.

The topic I found most interesting and useful was **subqueries in SQL**, as they allow complex, nested data retrieval that can simplify queries and reduce redundancy, making database operations both powerful and efficient.